

Results

Study 1: Feasibility of Predicting Keep/Remove Decisions

Motivation

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Formulating the task

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Dataset

Prolific participants (n=1,321) reviewed pairs of original and mirrored posts (n=959 pairs), resulting in a final dataset of n=13,250 rows, with each row representing a human keep/remove decision for the pair of posts. In these “linked fate” trials, the users saw both the original post and its mirrored equivalent, randomly ordered, and were asked to make a keep/remove decision for the pair. In this dataset, 66.2% of post pairs were kept (and 33.8% removed). We create training labels for each post by taking the modal label across all raters (average n=13.8 labels per post).

Training an initial set of models with explainable text features

We first generated a set of explainable text features:

- Does the text involve intergroup discussion?
- Does the text have PRIME content?
- Does the text have a positive valence?
- Does the text have toxic content?
- What is the political stance of the text?
- What is the number of characters in the text?
- What is the average sentence length?
- What is the Flesch-Kincaid reading score?

We then trained two models on these features: a logistic regression model and an XGBoost model. We report those results in a table

Caption: Model performance for predicting keep/remove decisions. Precision, recall, F1, and ROC-AUC are computed with the remove decision as the positive class ($y = 1$).

Model	Split	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (always predict keep)	Test	0.649	0.000	0.000	0.000	0.500
Logistic regression	Train	0.699	0.572	0.407	0.475	0.675
Logistic regression	Test	0.695	0.597	0.402	0.480	0.678
XGBoost	Train	0.737	0.667	0.430	0.523	0.777
XGBoost	Test	0.693	0.603	0.367	0.457	0.702

We notice a slight lift in the logistic regression and XGBoost models as compared to a naive baseline, driven by improvements in recall of remove decisions. We also observe likely overfitting on the XGBoost model given the sparsity of our n=959 dataset.

Study 2

In Study 2, ...

(initial results from Study 2)

Model performance

(model performance)

Scaling curves

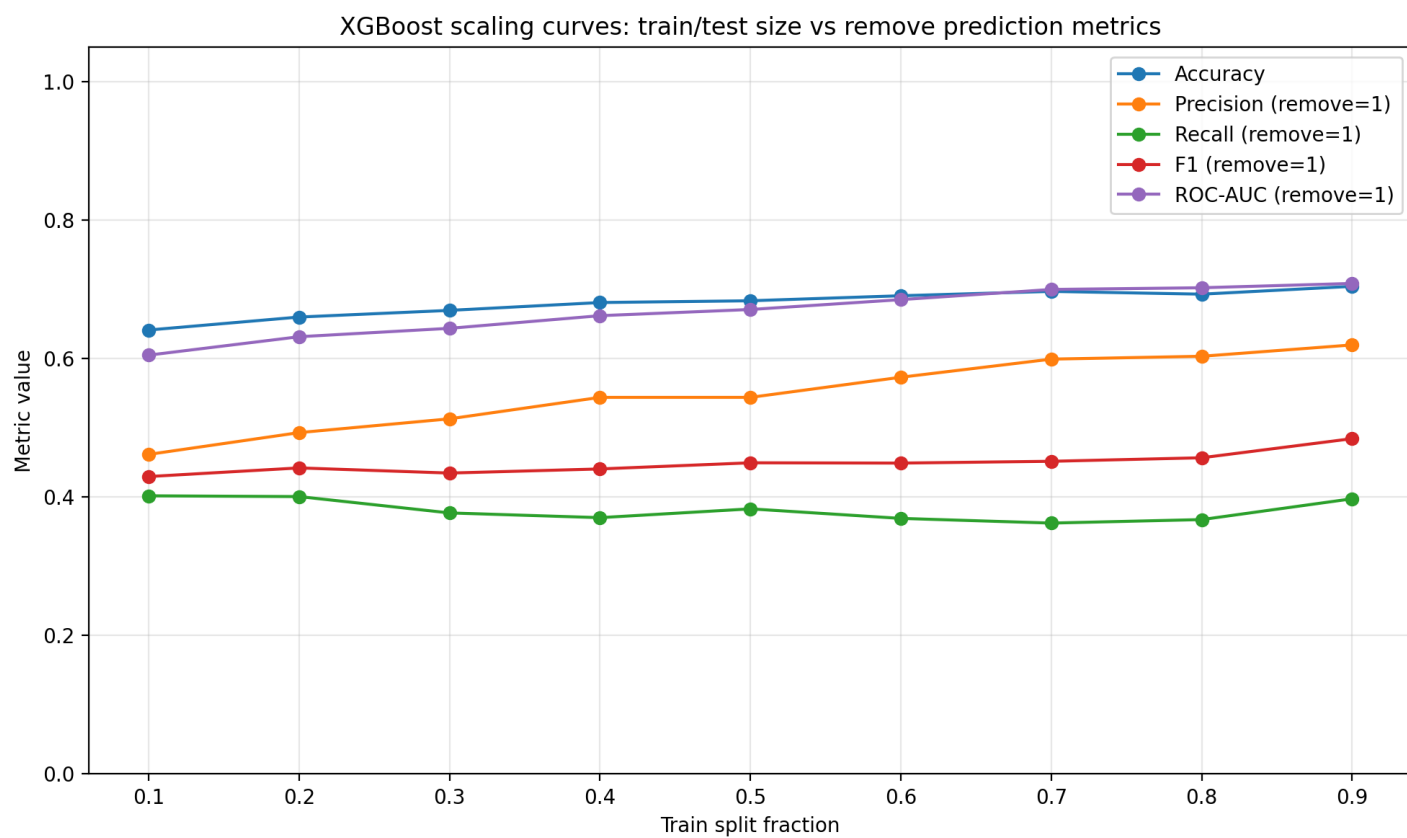


Figure 1: XGBoost scaling curves